

SANDEEP BANDI

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying machine learning systems that solve real business problems - from fraud detection in insurance to GL coding automation and GenAI-powered invoice matching. Experienced across the full ML lifecycle: data pipelines, model development, experiment tracking, and production deployment on cloud platforms like AWS and Azure. Skilled in deep learning, NLP, and LLM-based solutions using Azure AI Foundry and OpenAI APIs. Focused on writing clean, reproducible code and implementing robust MLOps practices (MLflow, Docker, CI/CD) to ensure models perform reliably in production.

TECHNICAL SKILLS

Languages & Frameworks: Python, R, SQL, Flask, FastAPI

ML & Deep Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, ANN, CNN, RNN

NLP & GenAI: Transformers, Hugging Face, LangChain, OpenAI APIs.

MLOps & CI/CD: MLflow, DVC, GitHub Actions, Jenkins, Apache Airflow, Docker, Kubernetes

Cloud Platforms: AWS, Azure, GCP

Databases: MySQL, PostgreSQL, MongoDB, SQL Server, AWS Glue

Visualization & BI: Power BI, Matplotlib, Seaborn

Dev Tools: Git, GitHub, VS Code, Jupyter Notebook

PROFESSIONAL EXPERIENCE

AI/ML Engineer

May 2025 – Present

PairSoft · Full-time · Remote

- **Papersave (GL Coding Automation):**
 - Built and deployed an end-to-end ML pipeline for multi-tenant GL account classification, serving 40+ enterprise tenants via FastAPI microservice with XGBoost, LightGBM, and TensorFlow models on Azure.
 - Engineered automated hyperparameter tuning using RandomizedSearchCV with stratified cross-validation, achieving 75% Top-1 and 92% Top-3 accuracy across 125 GL account classes on 53K+ invoices.
 - Designed scalable multi-tenant architecture with tenant-specific model artifacts and categorical encoding mappings, enabling seamless onboarding of new clients with minimal configuration.
 - Implemented MLflow for experiment tracking, model versioning, and performance monitoring across multiple tenants.
 - Built FastAPI REST API with real-time inference endpoints for GL account prediction, serving top-3 predictions with confidence scores to PaperSave's document management platform.
- **Smart PO Matching & Intelligent Invoice Extraction (GenAI):**
 - Led development of GenAI-powered invoice automation platform - from data extraction to PO reconciliation - serving enterprise AP operations.
 - Architected scalable Azure-hosted APIs with Parquet-based caching for optimized PO lookups and real-time matching.
 - Owned API contract design, technical documentation, and partner enablement for seamless third-party integration.
 - Delivered production-ready solution with Oracle ERP integration and partner onboarding (PaperSave and Apro Customers).

Tech Stack: Python, Azure App Services, Azure AI Services, REST APIs, Parquet, Docker.

ML Engineer

Jun 2024 – Sep 2024

BeWorld App · Part-time · Hyderabad, India

- Designed and deployed a personalized feed ranking system using hybrid recommendation (collaborative + content-based filtering) for social media engagement.
- Built ML models (Logistic Regression, XGBoost) to predict post interest scores using MongoDB-sourced engagement and metadata features.
- Served ranked content via FastAPI endpoints and deployed scalable services on AWS EC2.

- Implemented CI/CD pipelines with Docker and Jenkins for reliable, automated model deployment.
- Collaborated with product and backend teams for seamless integration into the production app.

System Engineer

Jan 2022 – Feb 2024

Tata Consultancy Services · Full-time · Hyderabad, India

- Built and deployed ML models (Scikit-learn, XGBoost) for risk scoring and fraud detection in insurance workflows.
- Designed automated data pipelines for claims and policyholder data processing, improving reporting accuracy and efficiency.
- Conducted EDA, feature engineering, and validation to ensure model quality in production environments.
- Deployed solutions on AWS EC2 and integrated into enterprise batch-processing pipelines.
- Collaborated across data engineering, business, and cloud teams to deliver actionable ML-powered tools.

Tech Stack: Python, Pandas, Scikit-learn, XGBoost, AWS EC2, SQL, Jupyter, Git

Project Intern

Nov 2021 – Feb 2022

iNeuron.ai · Remote

- Developed a machine learning system for income prediction through a complete end-to-end ML pipeline.
- Executed all pipeline stages including data ingestion, EDA, transformation, model training, and deployment.
- Built and containerized the solution using Python and Docker; deployed on AWS for real-time inference.

PROJECTS

Hotel Reservation Cancellation Prediction | Python, MLflow, Jenkins, Docker, Flask

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- Built an end-to-end ML solution to predict hotel reservation cancellations, enabling proactive customer engagement and revenue optimization.
- Implemented data versioning (DVC), experiment tracking (MLflow, CometML), and automated CI/CD deployment (Jenkins, Docker, GCP Cloud Run).

Hybrid Anime Recommendation System | Python, Scikit-learn, Flask, Streamlit

[GitHub](#) | [Live App](#)

- Developed a hybrid recommendation system combining collaborative and content-based filtering for personalized anime suggestions.
- Designed a modular ML pipeline with Flask backend and Streamlit front-end for deployment-ready application.

Adult Census Income Prediction | Python, Flask, Scikit-learn, MongoDB, AWS

[GitHub](#) | [Live App](#)

- Designed a binary classification system (Logistic Regression, 85% accuracy) to predict income levels with a modular data processing pipeline.
- Built a web application for real-time predictions with integrated preprocessing for numeric and categorical features.

Gemstone Price Prediction | Python, LightGBM, XGBoost, Scikit-learn

[GitHub](#) | [Live App](#)

- Built and deployed a regression model using cross-validation and ensemble methods to estimate gemstone prices.
- Analyzed and visualized key pricing factors using Pandas and Matplotlib for interpretable model outputs.

EDUCATION

Bachelor of Technology in Computer Science Engineering

2016 – 2020

Lovely Professional University · Punjab, India

CERTIFICATIONS

- Full Stack Data Science — iNeuron [View Certificate](#)
- Complete ML, DL, NLP Bootcamp — Udemy [View Certificate](#)
- Python — Kaggle [View Certificate](#)